



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1202 – Introduction to Probability Theory
Tutorial No: 3

1. There are 8 unrelated questions on a multiple-choice test. Each question has 5 options and only one is correct. What is the probability that, by guessing a student can answer:
 - a. Exactly three questions are correctly?
 - b. All questions are incorrectly?
 - c. At most two questions are correctly?
 - d. At least 2 but no more than 5 correctly?
 - e. Less than 2 or more than 5 correctly?
2. Experience has shown that 30% of all persons afflicted by a certain illness recover. A drug company has developed a new medication. Ten people with the illness were selected at random and received the medication; nine recovered shortly thereafter. Suppose that the medication was absolutely worthless. What is the probability that at least nine of ten receiving the medication will recover?
3. Suppose that a lot of 5000 electrical fuses contains 5% defectives. If a sample of 5 fuses is tested,
 - a. find the probability of observing at least one defective.
 - b. The large lot of electrical fuses of (part a) is supposed to contain only 5% defectives. If $n = 20$ fuses are randomly sampled from this lot, find the probability that at least four defectives will be observed.
4. Suppose that 20% of all copies of a particular textbook fail a certain binding strength test. Let X denote the number among 15 randomly selected copies that fail the test
 - a. What is the probability that at most 8 fail the test
 - b. What is probability that exactly 8 fails?
 - c. What is the probability that at least 8 fails?
 - d. What is the probability that between 4 and 7
5. If 75% of all purchases at a certain store are made with a credit card and X is the number among ten randomly selected purchases made with a credit card. Calculate value of the mean and standard deviation for this experiment.

6. If a publisher of nontechnical books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is 0.005 and errors are independent from page to page:
 - a. what is the probability that one of its 400-page novels will contain exactly one page with errors?
 - b. What is the probability of at most three pages with errors?
7. Customers arrive at a restaurant at a rate of 10 customers per hour. Assuming that the number of arrivals follows a Poisson distribution, find the probability that more than 14 customers arrive within a particular hour.
8. A manufacturer produces light-bulbs that are packed into boxes of 100. If quality control studies indicate that 0.5% of the light-bulbs produced are defective, what percentage of the boxes will contain:
 - a. no defective?
 - b. 2 or more defectives?
9. Telephone calls enter the support desk of a company on the average of two every 3 minutes. If one assumes an approximate Poisson process, what is the probability of,
 - a. 0 calls in a 9 minutes period
 - b. Exactly 4 calls in a 9-minute period
 - c. 4 or less calls during a nine-minute period
 - d. What are the maximum calls that should enter 9 min with 90% certainty?

(Please answer question number 1, 2, 4, 5, 6, 8 and 9 and submit the answer script to the tutorial room on or before 13th of December 2023)